

# Statistical Foundations of Pairs Trading via Cointegration

## Conceptual setup for pairs trading with price series

In the “classic” pairs trading formulation, you are not trying to predict the direction of each asset’s price. You are trying to find a **linear combination of two price series that is mean-reverting** (stationary), so that short-run deviations can be traded as temporary mispricings. This is exactly the *cointegration* idea: two series can be individually nonstationary (typically modeled as having a unit root / being integrated of order 1), yet some linear combination of them is stationary. <sup>1</sup>

Formally, let  $y_t$  and  $x_t$  denote (log-)prices of two assets. A bivariate cointegration relationship means:

- $y_t$  is  $I(1)$  and  $x_t$  is  $I(1)$  (each becomes stationary after first differencing), but
- there exists  $\beta$  (and possibly an intercept and/or trend) such that the **spread**

$$u_t = y_t - \beta x_t \quad (\text{or } u_t = y_t - \alpha - \beta x_t)$$

is  $I(0)$  (stationary). <sup>2</sup>

The economic/stat-arb intuition used in pairs trading then becomes: if  $u_t$  is stationary, deviations of  $u_t$  from its mean are *bounded in distribution* and tend to revert, so trading rules based on  $u_t$  (or its z-score) have a statistical basis that simple co-movement metrics do not. <sup>2</sup>

A practical detail that matters for *your* task: many quant treatments explicitly distinguish **cointegration on (log-)prices** vs **correlation on (log-)returns**, because they describe different horizons (long-run equilibrium vs short-run co-movement). <sup>3</sup>

## Correlation vs cointegration in a way that prevents common mistakes

**Correlation** (e.g., Pearson correlation) is a *contemporaneous linear association* measure. In finance it is typically applied to **returns** (or log-returns) to describe short-horizon co-movements. <sup>3</sup>

**Cointegration** is a *time-series equilibrium* concept: it is about whether two **nonstationary levels** share a **common stochastic trend** such that a particular linear combination is stationary (mean-reverting). <sup>4</sup>

A useful “guardrail” distinction (often used in quant lectures) is:

- Correlation of  $\Delta y_t$  (returns)  $\approx$  **short-term co-movement**
- Cointegration of  $y_t$  (log-prices)  $\approx$  **long-term co-movement / equilibrium constraint** <sup>3</sup>

This distinction matters because you can have:

- **High return correlation but no cointegration** (assets move together short-run but spread drifts over time), and also
- **Cointegration with modest return correlation** (long-run equilibrium exists, but short-run idiosyncratic noise weakens contemporaneous return correlation). <sup>5</sup>

## Why “high correlation” is not enough to select a pair

Pairs trading needs a **mean-reverting spread**, not just “they move together a lot.” Correlation alone can fail for three reasons that show up repeatedly in econometrics:

First, with trending/nonstationary series, high fit measures (and by extension, “strong association” signals) can be **spurious**. The classic warning is that regressions among unrelated nonstationary series can look highly significant (high  $R^2$ , strong t-stats) even when there is no meaningful stable relationship. <sup>6</sup>

Second, correlation does **not** encode the key condition you need for pairs trading: that some  $u_t = y_t - \beta x_t$  is  $I(0)$ . Two prices can be highly correlated while their difference (or hedge-ratio spread) behaves like a random walk (nonstationary), meaning deviations are *not* expected to revert. <sup>4</sup>

Third, correlation is a **single-number summary** of co-movement at (implicitly) one horizon. Cointegration is explicitly about **long-run behavior** (shared stochastic trend + stationary deviation). That long-run property is what links directly to the “convergence” premise of pairs trading. <sup>4</sup>

A concise way to say it in a “Pairs Selection Report” is:

High correlation may indicate co-movement, but only cointegration (via residual stationarity) provides statistical evidence that a spread is mean-reverting rather than drifting. <sup>7</sup>

## Engle–Granger two-step method and what you actually run the ADF test on

The bivariate workflow most aligned with a pairs task is the **Engle–Granger two-step** approach, developed by Robert F. Engle <sup>8</sup> and C. W. J. Granger <sup>9</sup>. In its common operational form:

### Step one (estimate the long-run relationship).

Run the cointegrating regression in levels (often log-prices):

$$y_t = \alpha + \beta x_t + \varepsilon_t$$

where  $\hat{\varepsilon}_t$  is interpreted as the deviation from equilibrium (the spread). <sup>10</sup>

### Step two (test whether the deviation is stationary).

Run a unit-root test (often an ADF-type test) on the **estimated residual series**  $\hat{\varepsilon}_t$ . The Engle–Granger logic is exactly: “cointegration exists if the residual is  $I(0)$ .” <sup>10</sup>

This answers the most important implementation question in your list:

### What does the ADF test get run on, and why?

In the Engle–Granger procedure, the unit-root test is run on the **cointegrating residual** (your candidate spread), not on the individual price series—and not on returns—because the definition of cointegration is “a linear combination is stationary.” <sup>10</sup>

In `statsmodels`, `tsa.stattools.coint` explicitly implements the “augmented Engle–Granger” residual-based test; it states that the null is **no cointegration**, assumes the inputs are  $I(1)$ , and returns “the t-statistic of unit-root test on residuals” plus MacKinnon-based p-values/critical values. <sup>11</sup>

A technical detail that prevents a common mistake: within the `statsmodels` implementation, the ADF regression in step two is called on residuals with **no constant/trend** (`regression="n"`), consistent with the idea that deterministic components were already handled in step one’s cointegrating regression. <sup>12</sup>

### Why you cannot treat this as “just an ordinary ADF on any series”

Even though the mechanics look like an ADF test, the **sampling distribution is not the standard Dickey–Fuller table** you might use for a raw series, because the residuals are *generated regressors* (estimated  $\beta$  means the test statistic’s distribution changes). Econometrics notes emphasize that you need **cointegration-specific critical values** (often called CADF), not the vanilla DF/ADF critical values. <sup>13</sup>

That is why packages typically use MacKinnon-style response-surface approximations (step-two p-values/critical values tailored to cointegration testing), credited to James G. MacKinnon <sup>14</sup> and updated over time. <sup>15</sup>

## Residual stationarity and its meaning for long-run equilibrium

The equilibrium interpretation is not an informal trading story—it is part of the original econometric framing. The Engle–Granger paper explicitly interprets  $\alpha'x_t = 0$  as a *long-run equilibrium* and states that cointegration implies deviations from equilibrium are stationary with finite variance, even though the individual series are nonstationary. <sup>16</sup>

Two consequences are especially relevant for pairs trading:

If  $u_t$  is stationary, then “distance” away from equilibrium does not explode over time in distribution; it fluctuates around a stable mean and variance. This is the mathematical justification for building spread-based signals like z-scores on  $u_t$ . <sup>17</sup>

Cointegration implies an **error-correction representation**: cointegrated series can be written so that short-run changes respond to the lagged deviation  $u_{t-1}$ . The Engle–Granger discussion ties cointegration and error-correction models together (Granger representation theorem and its consequences). <sup>18</sup>

For a pairs trading report, you can express this connection compactly as:

$$\Delta y_t = \lambda (y_{t-1} - \alpha - \beta x_{t-1}) + \text{short-run dynamics} + \eta_t,$$

where  $(y_{t-1} - \alpha - \beta x_{t-1})$  is the error-correction term (the spread), and  $\lambda$  captures the speed of adjustment back toward equilibrium. Cointegration is the condition that makes this equilibrium-correction formulation coherent in levels. <sup>19</sup>

## When it is reasonable to conclude a pair is cointegrated in practice

A defensible (and assignment-friendly) standard for declaring a pair cointegrated under Engle–Granger is:

You have evidence that each price series is plausibly  $I(1)$  (or at least nonstationary in levels), because cointegration testing is posed as a relationship among  $I(1)$  variables; `statsmodels.coint` explicitly assumes this. <sup>20</sup>

Your residual-based test rejects the **null of no cointegration** at a pre-specified significance level (e.g., 5%), using the correct cointegration p-values/critical values (MacKinnon-style). In `statsmodels.coint`, this corresponds to a small p-value (and the function's documentation reiterates the null/alternative structure). <sup>21</sup>

You used a coherent deterministic specification (constant and/or trend) in the cointegrating regression, because deterministic terms affect what “stationary” means and alter valid testing; econometrics notes explicitly caution that unit root tests should account for deterministic components appropriately. <sup>22</sup>

For your setting (about **500 pairs**), it is also statistically responsible to acknowledge the multiple testing problem: if you test many pairs at  $\alpha = 0.05$ , you should expect some false “discoveries” even if none are truly cointegrated. A common remedy is controlling the **false discovery rate (FDR)** using the Yoav Benjamini <sup>23</sup>–Yosef Hochberg <sup>24</sup> procedure, which compares sorted p-values  $p_{(i)}$  to thresholds of the form  $(i/m)q$  and rejects up to the largest satisfying index. <sup>25</sup>

Finally, a pragmatic “cointegration is plausible” claim often includes a *stability* check: the spread should be stationary not only in the full sample but also across subwindows (cointegration can change with regime). This is not “proved” by Engle–Granger alone, but it is consistent with the idea that cointegration encodes a long-run relationship that can be disrupted by structural change. <sup>26</sup>

## Frequent pitfalls for beginners, especially with minute-level prices and `statsmodels`

Several errors come up repeatedly in pairs-trading implementations; the ones below are particularly likely given your constraints (minute data, many pairs, ADF usage):

Mistaking correlation for cointegration: computing correlation (often on returns) and then concluding there is a tradable mean-reverting spread. Correlation is short-horizon co-movement; cointegration is long-run equilibrium in levels. <sup>27</sup>

Running the ADF test on the wrong object: a common error is running ADF on price levels (which only tests whether a single series is stationary) or on returns (which are often stationary by construction), and then asserting cointegration. The Engle-Granger logic requires the unit-root test to be run on the **residual/spread** from the cointegrating regression. <sup>28</sup>

Using the wrong critical values / p-values for the residual-based test: because the residual is estimated, the distribution differs from the standard DF/ADF tables; you need cointegration-specific critical values, which is why software uses MacKinnon approximations and warns about distribution differences. <sup>29</sup>

Ignoring deterministic components: mixing “with constant” vs “no constant” decisions inconsistently between step one and step two can change conclusions. `statsmodels.coint` includes trend options in step one and then applies an ADF regression without constant on the residual; this is aligned with the idea that the residual has already been de-measured/detrended. <sup>30</sup>

Autolag and dependence issues: ADF-type tests require lag selection to control serial correlation in the test regression; both `statsmodels.adfuller` and the Stata documentation highlight that lagged differences are included to address serial correlation, and the choice of deterministic terms changes the null model. <sup>31</sup>

Multiple testing blindness: with ~500 pairs, “significant at 5%” without adjustment is often too optimistic. FDR control is a standard statistical response when you are screening many hypotheses. <sup>25</sup>

High-frequency data complications that contaminate inference: minute-level prices are affected by market microstructure noise and data-handling issues (bid-ask bounce, price discreteness, idleness/stale prices, asynchronous trading). High-frequency econometrics explicitly models observed prices as containing a microstructure noise component, and empirical work emphasizes that cointegration relationships in high-frequency settings can be challenging and time-varying (including intraday effects). <sup>32</sup>

Near-collinearity and numerical instability: cointegrating regressions can become numerically unstable if two series are almost perfectly collinear; `statsmodels` explicitly warns that the cointegration test is not reliable in that edge case and may return extreme values. <sup>33</sup>

## Summary for your Pairs Selection Report

### Methodology note

Use (log-)price series  $y_t, x_t$  sampled on identical minute timestamps, then test whether a linear combination  $u_t = y_t - \alpha - \beta x_t$  is stationary even when  $y_t$  and  $x_t$  are nonstationary. Cointegration is defined as the existence of such an  $I(0)$  combination and is interpreted as a long-run equilibrium with stationary deviations. <sup>4</sup>

Implement cointegration screening pair-by-pair via the Engle-Granger two-step method: estimate  $\alpha, \beta$  by OLS in the cointegrating regression (step one), compute residuals  $\hat{u}_t$ , then apply a residual-based unit-root test (ADF-type) to  $\hat{u}_t$  (step two). A pair is supported as cointegrated if you reject the null of no cointegration using cointegration-appropriate critical values/p-values (MacKinnon-style), as done in `statsmodels.tsa.stattools.coint`. <sup>34</sup>

Because you are screening around 500 pairs, report how you address multiplicity (e.g., Benjamini–Hochberg FDR) so that your “selected” set is not dominated by false positives. <sup>25</sup>

## Checklist: “must get right”

- Use the right objects: cointegration is about **(log-)prices/levels**, while correlation is typically about **(log-)returns**. <sup>5</sup>
- Align timestamps and market hours so  $x_t$  and  $y_t$  are truly synchronous minute series; otherwise microstructure/lead-lag effects can distort results. <sup>35</sup>
- Step one: estimate the cointegrating regression in levels (often with a constant). <sup>36</sup>
- Step two: run the unit-root test on the **cointegrating residual/spread**  $\hat{u}_t$ , not on returns and not on the raw prices. <sup>37</sup>
- Use cointegration-appropriate inference (MacKinnon-style p-values/critical values), not “plain DF tables,” because the residual-based test has a different distribution. <sup>38</sup>
- State the hypothesis clearly: for `statsmodels.coint`, the **null is no cointegration**; selection corresponds to rejecting this null at your chosen level. <sup>21</sup>
- Address multiple testing across ~500 pairs (e.g., BH-FDR) and report the control rule you used. <sup>25</sup>
- Flag (or exclude) pathological near-collinearity cases where the test becomes numerically unreliable. <sup>33</sup>

## Checklist: common misconceptions to avoid

- “Correlation  $\approx$  cointegration”: correlation measures short-horizon co-movement; cointegration is a long-run equilibrium property of levels. <sup>27</sup>
- “A high correlation is enough for a pairs trade”: high association can be spurious with nonstationary series; what you need is a stationary spread. <sup>39</sup>
- “ADF on returns proves cointegration”: returns can be stationary even when levels are not; cointegration requires stationarity of a *linear combination of levels*. <sup>40</sup>
- “ADF on each price series is the cointegration test”: ADF on a single series only addresses that series’ unit root; Engle–Granger tests cointegration by testing the residual for a unit root. <sup>41</sup>
- “Any ADF p-value works”: residual-based cointegration testing needs cointegration-specific distributions/critical values (MacKinnon/CADF). <sup>29</sup>
- “Minute data is just ‘more data’”: at high frequency, microstructure noise, stale prices, and intraday effects can change inference and even the apparent existence of cointegration. <sup>32</sup>

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<sup>1</sup> <sup>2</sup> <sup>4</sup> <sup>7</sup> <sup>16</sup> <sup>17</sup> <sup>18</sup> <sup>19</sup> <sup>26</sup> Co-Integration and Error Correction: Representation, Estimation, and Testing

<https://nomanarshed.wordpress.com/wp-content/uploads/2013/10/engle-granger.pdf>

<sup>3</sup> <sup>5</sup> <sup>8</sup> <sup>23</sup> <sup>27</sup> Pairs Trading

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<sup>6</sup> <sup>39</sup> Spurious regressions in econometrics

<https://gwern.net/doc/economics/1974-granger.pdf>

<sup>9</sup> <sup>10</sup> <sup>22</sup> <sup>28</sup> <sup>34</sup> <sup>36</sup> <sup>37</sup> <sup>40</sup> fmmaster.dvi

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